

Polynomial Functors and Evolutionary Operators

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Abstract

We establish a precise connection between polynomial functors in the sense of Gambino-Kock and evolutionary operators in genetic programming. Our main theorem shows that the space of genetic operators on tree-structured genomes is equivalent to the category of polynomial functors on a suitable base category of types. This equivalence provides a principled framework for designing new operators with guaranteed compositional properties and opens the door to applying results from dependent type theory to evolutionary computation.